

Explicit Cubic Threefolds Rational after Two Stabilizations

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Abstract

We exhibit explicit smooth cubic threefolds $X/\mathbf{Q}$ for which $X \times \mathbf{P}^1$ is irrational but $X \times \mathbf{P}^2$ is rational. Thus the least number of independent variables required to make these threefolds rational is exactly two, over both $\mathbf{Q}$ and $\mathbf{C}$. The rationality statement follows from a general result for quartic del Pezzo surfaces: if k has characteristic zero, $S(k) \neq \emptyset$, and $\text{Pic}(S_k)$ is a stably permutation Galois module, then $S \times \mathbf{A}^2$ is k -rational. Its proof identifies a rational quotient of the projective Cox model by a saturated rank-three subtorus; the residual two-dimensional torus is rational. Applied to the quartic-del-Pezzo fibrations of Tschinkel and Zhang, this proves rationality after two stabilizations for both of their cubic families. Their stable-rationality conclusions therefore hold with the uniform bound $\mathbf{P}^2$. As a consequence, there are smooth projective nonrational fourfolds $Y/\mathbf{Q}$ such that $Y \times \mathbf{A}^1$ is rational.

Keywords. cubic threefold; quartic del Pezzo surface; stable rationality; universal torus; variety with one apparent double point; algebraic torus; rationality after affine-line stabilization.

1 Introduction and main results

For a stably rational but nonrational variety, one can ask for the least number of independent variables that makes its function field rational. The question is especially sharp for cubic threefolds. Every smooth complex cubic threefold remains irrational after multiplication by $\mathbf{P}^1$ [3, Theorem 1.1], while explicit smooth cubic threefolds over $\mathbf{Q}$ are now known to be stably rational [4, Theorem 1.3]. We determine the least number for two such threefolds.

For a projective variety V/k , put

$$\ell_k(V) = \min\{m \geq 0 : V \times \mathbf{A}^m \text{ is } k\text{-rational}\}.$$

We set $\ell_k(V) = \infty$ if the set is empty. Since $\mathbf{A}^m$ and $\mathbf{P}^m$ are birational, one obtains the same value with $\mathbf{P}^m$ in place of $\mathbf{A}^m$. Tschinkel and Zhang consider this minimum in their discussion of levels of stable rationality [4, p. 13].

Consider the smooth cubic threefolds

$$X_1 : (x_4 - 2x_3)x_1^2 + 3(x_4 + 2x_3)x_2^2 + 3x_3^2x_4 - x_4^3 + x_5^3 = 0, \quad (1)$$

$$X_3 : x_4(x_1^2 + 2x_1x_2) + x_3(x_1^2 + x_1x_2 + x_2^2) + x_3^3 - x_3x_4^2 + x_4^3 + 2x_5^3 = 0. \quad (2)$$

Theorem 1.1 (Exact level for the cubic threefolds). *For $j \in \{1, 3\}$,*

$$\ell_{\mathbf{Q}}(X_j) = \ell_{\mathbf{C}}((X_j)_{\mathbf{C}}) = 2.$$

Equivalently, $(X_j)_{\mathbf{C}} \times \mathbf{P}^1$ is irrational, whereas $X_j \times \mathbf{P}^2$ is rational over $\mathbf{Q}$.

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The mechanism behind Theorem 1.1 is the following surface theorem.

Theorem 1.2 (Two-dimensional stabilization). *Let k be a field of characteristic zero and let S/k be a smooth quartic del Pezzo surface. Suppose that $S(k) \neq \emptyset$ and that $\text{Pic}(S_{\bar{k}})$ is a stably permutation $\text{Gal}(\bar{k}/k)$ -module. Then*

$$S \times \mathbf{A}^2 \quad \text{and, equivalently,} \quad S \times \mathbf{P}^2$$

are k -rational. In particular, this holds for every stably k -rational smooth quartic del Pezzo surface.

The proof combines the Cox model, tangent-projection theorem, Galois-type classification, and generic splitting of the universal torsor developed in [4] with the rank-three quotient constructed below. Under the rational-point hypothesis, condition (1) of [4, Corollary 4.3] is exactly the stably permutation hypothesis in Theorem 1.2. Theorem 1.2 proves that this condition implies rationality after two variables; Manin's necessary condition gives the converse from stable rationality, while [4, Proposition 4.1 and Lemma 4.2] identify the other equivalent conditions in that corollary. Thus its equivalence holds with the positive direction quantified by $S \times \mathbf{A}^2$ rational. The stable-permutation resolution in [4, p. 13] gives the uniform bound eleven; the quotient construction here gives two.

For $r \geq 0$, let $X_{j,r} \subset \mathbf{P}_{\mathbf{Q}}^{r+4}$ be obtained from equation (1) or (2) by adding $w_1^3 + \cdots + w_r^3$ to its left-hand side. Thus $X_{j,0} = X_j$.

Corollary 1.3 (Two families of cubic hypersurfaces). *For every $r \geq 0$ and $j \in \{1, 3\}$, the variety $X_{j,r} \times \mathbf{P}^2$ is rational over $\mathbf{Q}$.*

The smoothness, fibrations, and generic-fibre Galois calculations in [4, Propositions 5.1–5.2] are the inputs to Corollary 1.3. Consequently the stable-rationality conclusions of those propositions, and hence the examples in [4, Theorem 1.3], hold with the uniform bound $X_{j,r} \times \mathbf{P}^2$ rational.

Corollary 1.4 (Rationality after affine-line stabilization). *For $j \in \{1, 3\}$, the variety $Y_j = X_j \times \mathbf{P}^1$ is a smooth projective fourfold over $\mathbf{Q}$ which remains nonrational over $\mathbf{C}$, while $Y_j \times \mathbf{A}^1$ is rational over $\mathbf{Q}$.*

This is a statement about birational rationality after taking a product with $\mathbf{A}^1$. It is not a counterexample to the Zariski cancellation problem, which concerns isomorphisms between affine cylinders.

Corollary 1.5 (Variation in the moduli of cubic threefolds). *On the moduli space of smooth complex cubic threefolds, $\ell_{\mathbf{C}}$ is equal to two at the two displayed points and is infinite at a very general point.*

2 A criterion for rational torus quotients

Let $Z \subset \mathbf{P}(V)$ be a geometrically integral n -fold over an infinite field k . At a smooth point $p \in Z(k)$, write

$$\pi_p : Z \dashrightarrow \mathbf{P}(V/\widehat{T}_p Z) \tag{3}$$

for projection from the embedded tangent space; the hat denotes the affine cone. In the application, $Z \subset \mathbf{P}^{2n+1}$ has one apparent double point, and π_p is birational for general p by [4, Theorem 2.4].

Theorem 2.1 (A quotient criterion). *Let a rank- r k -torus T act projectively linearly and generically freely on $Z \subset \mathbf{P}(V)$, and suppose that the generic tangent projection π_p is birational to $\mathbf{P}^n$. Choose a diagonal linear lift of the action over k^s . Its weights are determined up to a common translation. Suppose that it has nonzero weight spaces $V_{w_0}, \dots, V_{w_r}$ such that $w_1 - w_0, \dots, w_r - w_0$ is a $\mathbf{Z}$ -basis of $X^*(T_{k^s})$. Assume that the sum $W = V_{w_0} \oplus \dots \oplus V_{w_r}$ and the sum $\widehat{B}$ of all other weight spaces descend to complementary k -subspaces of V ; write $B = \mathbf{P}(\widehat{B})$.*

Suppose that, for a general $p \in Z(k)$, there is a k -defined codimension- r linear subspace

$$\Lambda \supset \langle T_p Z, B \rangle$$

with equations $\lambda_1, \dots, \lambda_r$ such that, at a general $q \in Z$, the matrix

$$A(q)_{ij} = \lambda_i(q_j), \quad q_j \in V_{w_j}, \quad (4)$$

has rank r and all $r + 1$ maximal minors are nonzero. Then the Rosenlicht quotient Z/T is k -rational.

Proof. The proof has two parts. The maximal minors produce a unique point of Λ on a general T -orbit. The tangent projection then identifies that rational section with projective space.

The image of Λ under π_p is a linear $\mathbf{P}^{n-r} \subset \mathbf{P}^n$, and

$$\pi_p|_{\Lambda \cap Z} : \Lambda \cap Z \dashrightarrow \mathbf{P}^{n-r} \quad (5)$$

is birational on the open where π_p is an isomorphism. It remains to show that $\Lambda \cap Z$ is a rational section of the T -action.

For a geometric point $q = \sum q_j + q_B$, where $q_j \in V_{w_j}$ and $q_B \in \widehat{B}$, define

$$\kappa_j(q) = (-1)^j \det A(q)_{\widehat{j}}, \quad 0 \leq j \leq r. \quad (6)$$

The equations of Λ vanish on B , so $tq \in \Lambda$ is equivalent to

$$A(q)(\chi^{w_0}(t), \dots, \chi^{w_r}(t))^t = 0.$$

Cramer's identity says that the kernel is generated by $(\kappa_0(q), \dots, \kappa_r(q))$. The basis hypothesis gives a unique $t(q) \in T(k^s)$, in projective coordinates, through

$$\chi^{w_j - w_0}(t(q)) = \frac{\kappa_j(q)}{\kappa_0(q)}, \quad 1 \leq j \leq r. \quad (7)$$

These are Laurent monomials in the displayed ratios. The construction is unchanged if all weights are translated by one character, since only their differences occur. It is also independent of the ordering of the selected weight spaces: intrinsically, $s(q) = t(q)q$ is the unique point of $Tq \cap \Lambda$. Because W , $\widehat{B}$, and Λ descend, this uniqueness makes s Galois equivariant. Thus s is a k -rational, orbit-invariant retraction to $\Lambda \cap Z$ on the open where the minors do not vanish. Hence $\Lambda \cap Z \dashrightarrow Z/T$ is birational.

We record the inverse and its domain. Choose linear forms $\rho_0, \dots, \rho_{n-r}$ whose restrictions give homogeneous coordinates on the image of Λ in $\mathbf{P}(V/\widehat{T}_p Z)$. Let $U_p \subset Z$ be an open on which π_p is an isomorphism, and let $U \subset U_p$ be the further open where all κ_j , $\rho_0(s(q))$ is nonzero and the action is free. Then

$$\Phi : U/T \longrightarrow \mathbf{P}^{n-r}, \quad [q] \longmapsto [\rho_0(s(q)) : \dots : \rho_{n-r}(s(q))] \quad (8)$$

is the quotient parametrization. If $R_\alpha(x)$ are homogeneous equations for Z , the graph of its inverse over $\Phi(U/T)$ is the unique component dominating that open of

$$\begin{aligned} R_\alpha(x) &= 0 \quad \text{for every } \alpha, \\ \lambda_i(x) &= 0 \quad 1 \leq i \leq r, \\ y_j \rho_0(x) - y_0 \rho_j(x) &= 0 \quad 1 \leq j \leq n - r. \end{aligned} \tag{9}$$

Write its unique point over y as $x(y)$ and set $\Theta(y) = [x(y)]$. The last line gives $\Phi(\Theta(y)) = y$. Conversely, $s(q) \in \Lambda \cap U_p$ and has the same tangent-projection coordinates as $x(\Phi([q]))$; injectivity of $\pi_p|_{U_p}$ gives

$$\Theta(\Phi([q])) = [s(q)] = [q] \quad \text{in } U/T.$$

Thus both composites are the identity on the stated dense opens. In the Cox application, the R_α in (9) are the twenty quadrics in [4, Section 3]. Equations (7) and (9) give the rational map and its inverse without an unrecorded finite cover. $\square$

The integral-basis condition is essential. If the weight differences span a proper finite-index sublattice, the same equations generally define a finite cover of a T -orbit rather than a rational section.

The proof is constructive once equations for Z , a tangent point, and the linear forms λ_i are given. The signed minors in (6) compute the orbit correction, and (9) recovers the inverse parametrization by elimination. Thus a tangent-section witness produces explicit rational maps in either direction. No estimate for the complexity of that elimination is asserted.

3 The rank-three subtorus of the projective Cox model

Let T be the Néron–Severi torus, so $X^*(T_{\bar{k}}) = \text{Pic}(S_{\bar{k}})$. The Galois-invariant cocharacter

$$\lambda_{-K} : \mathbf{G}_m \longrightarrow T, \quad D \longmapsto (-K_S, D), \tag{10}$$

acts by simultaneous scalar multiplication on the sixteen Cox generators. Write $T_0 = T/\lambda_{-K}(\mathbf{G}_m)$, a torus of rank five. Over $\bar{k}$, the projective Cox model

$$Z = \text{Proj Cox}(S) \subset \mathbf{P}^{15}$$

is a seven-dimensional variety with one apparent double point [4, Lemma 3.2].

Use the marking of [4, Sections 3–4]. For the type I_3 Galois image, the sublattice

$$N_3 = \mathbf{Z}e_3 \oplus \mathbf{Z}e_4 \oplus \mathbf{Z}e_5 \subset X_*(T_{0,\bar{k}}) \tag{11}$$

is saturated and Galois stable. The two generators act on this basis by

$$\begin{pmatrix} -1 & 0 & 0 \\ -1 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & -1 \end{pmatrix}.$$

Let $T_3 \subset T_0$ be the corresponding rank-three subtorus. Four blocks of Cox generators have the following projective T_3 -weights:

weight	Cox generators
$(0, 1, 1)$	L_{13}, L_{23}, L_{35}
$(1, 0, 1)$	L_{14}, L_{24}, L_{45}
$(1, 1, 0)$	E_1, E_2, E_5
$(1, 1, 1)$	L_{12}, L_{15}, L_{25}

The four blocks form a Galois-stable set, but they are not individually Galois stable. On their displayed weights, the first generator acts by

$$(011\ 111)(101\ 110),$$

and the second by $(110\ 111)$, fixing 011 and 101. These are affine actions on projective weights: a different diagonal lift translates all four weights by a common character. Every other Cox generator lies in the Galois-stable subspace

$$B = \mathbf{P}\langle E_3, E_4, L_{34}, Q \rangle. \quad (12)$$

The determinant of the three weight differences from $(0, 1, 1)$ is one. Consequently the direct sum of the four selected blocks and the affine cone over B are complementary subspaces defined over the ground field, as required in Theorem 2.1.

Remark 3.1 (Saturation and the projective kernel). In the type- I_1 description, the three visible sign cocharacters in the last three coordinates are

$$(0, 1, 1), \quad (1, 0, 1), \quad (1, 1, 0).$$

They span the even-sum sublattice of $\mathbf{Z}^3$, of index two. Its saturation is $\mathbf{Z}^3$, obtained by adjoining

$$(1, 1, 1) = \frac{1}{2}((0, 1, 1) + (1, 0, 1) + (1, 1, 0)).$$

The map from the three parameter copies of $\mathbf{G}_m$ to their image in the projective torus has kernel μ_2 . The degree two obtained from the unsaturated parameters counts this kernel; it is not the degree of the Rosenlicht quotient. Saturation gives the integral-basis condition in Theorem 2.1.

Proposition 3.2 (A tangent section for the four Galois types). *For a smooth geometric quartic del Pezzo Cox model, there is a general tangent point p and a codimension-three linear subspace $\Lambda \supset \langle T_p Z, B \rangle$ which satisfies the maximal-minor condition of Theorem 2.1 for T_3 . If the Galois image has type I_0, I_1, I_2 , or I_3 , such a pair (p, Λ) exists over the ground field.*

Proof. Use the standard blowup points

$$(1 : 0 : 0), (0 : 1 : 0), (0 : 0 : 1), (1 : 1 : 1), (1 : a : b), \quad \Delta = ab(a-1)(b-1)(a-b) \neq 0. \quad (13)$$

For a tangent point with auxiliary coordinates $z = (z_1, z_2, z_3)$, the space of tangent hyperplanes containing B has dimension four. Evaluation on the four groups of Cox generators gives a 4-by-4 matrix. The following exact choices are used; $e = (e_1, \dots, e_5)$ and z' specify the second Cox point at which the evaluation matrix is tested.

i	z	e	z'
1	$(1, 3, 7)$	$(2, 3, 5, 7, 11)$	$(2, 4, 9)$
2	$(2, 5, 11)$	$(3, 4, 7, 13, 17)$	$(1, 6, 10)$
3	$(3, 8, 13)$	$(5, 7, 11, 17, 19)$	$(2, 9, 15)$
4	$(4, 9, 17)$	$(2, 5, 11, 19, 23)$	$(3, 10, 18)$

Let D_i be the determinant of this evaluation matrix, and let M_i be the fixed 8-by-8 Jacobian minor at the same tangent point. After discarding nonzero rational factors, their numerators are recorded in Appendix A. Witness i is valid precisely on the open $D_i M_i \neq 0$. The appendix proves, by eight localized cases and one explicit Bézout identity, that

$$\bigcup_{i=1}^4 \{D_i M_i \neq 0\} = \{\Delta \neq 0\}. \quad (14)$$

Thus every smooth geometric Cox model has a witness for which the tangent point is smooth and the four-block evaluation map is an isomorphism. A general three-frame in its four-dimensional source then gives a 3-by-4 matrix of rank three with all four maximal minors nonzero. In particular, the good incidence open used below is nonempty on every geometric fibre.

It remains to descend a good pair. Fix a twist of one of the four types. Because T_3 and B are Galois stable, tangent points, three-frames of hyperplanes containing $\langle T_p Z, B \rangle$, and orbit test points form a geometrically integral incidence space over k . More explicitly, after restricting to the open where $\dim \langle T_p Z, B \rangle$ is constant, the three-frames form an open in a relative frame bundle over the product of the smooth tangent-point locus with the orbit-test locus. Rank, nonvanishing, smoothness, and birational tangent projection define a nonempty Galois-invariant open in this bundle. Its projection to the tangent-point factor is dense constructible and therefore contains a nonempty open.

Choose the universal torsor whose fibre over a fixed point of $S(k)$ is trivial. Its k -points are Zariski dense by [4, Lemma 2.1], and the same holds in the projective Cox model. Hence the tangent-point open has a k -point. At that point, good three-frames form a nonempty open in an affine space over k , which has a k -point because k is infinite. No individual split-coordinate witness is asserted to descend.

For type I_3 , stability of N_3 , B , and the set of four selected Cox blocks is checked on the two displayed generators. Up to conjugacy in $W(D_5)$, the groups of types I_0, I_1, I_2 are subgroups of the group of type I_3 [4, Lemma 4.2]. Restriction and a change of marking preserve the same data. This exhausts the four Galois images allowed by the hereditary H^1 -condition; it is not a classification of arbitrary subgroups of $W(D_5)$. $\square$

4 Proofs of the rationality statements

Proof of Theorem 1.2. We first treat the minimal case by taking a quotient of the projective Cox model. An equivariant generic trivialization then identifies that quotient with the product of S and a two-dimensional torus. The nonminimal case reduces to the classical rationality theorem in degree at least five.

Suppose first that S is k -minimal. Since a stably permutation module has vanishing H^1 on every open subgroup, Proposition 4.1 and Lemma 4.2 of [4] place the Galois image in one of the four types used in Proposition 3.2. Lemma 3.2 and Theorem 2.4 of [4], together with Theorem 2.1, give

$$Z/T_3 \sim_k \mathbf{P}^4. \quad (15)$$

Let $\mathcal{T} \rightarrow S$ be a universal torsor under the Néron-Severi torus T , chosen to have a k -point above a fixed point of $S(k)$. The stable-permutation hypothesis and Hilbert's Theorem 90 give

$$H^1(F, T) = 0 \quad \text{for every field extension } F/k \quad (16)$$

by [4, Remark 2.2]. Thus the generic fibre over $k(S)$ has a point. Translation by the resulting rational section σ defines the T -equivariant birational map

$$S \times T \dashrightarrow \mathcal{T}, \quad (s, t) \mapsto \sigma(s)t. \quad (17)$$

The split scalar subtorus in (10) acts as common multiplication on every Cox coordinate. The universal torsor is a dense open in the affine Cox cone, and its scalar quotient is a dense open in Z . Quotienting (17) by that scalar therefore gives a T_0 -equivariant birational map $S \times T_0 \dashrightarrow Z$. Since $T_3 \subset T_0$ is a k -subtorus, Rosenlicht quotients give

$$Z/T_3 \sim_k S \times (T_0/T_3). \quad (18)$$

The quotient T_0/T_3 is a two-dimensional k -torus, hence is k -rational by [6]. Equations (15) and (18) prove that $S \times \mathbf{A}^2$ is k -rational.

If S is not k -minimal, successively contracting k -defined unions of pairwise disjoint exceptional curves gives a del Pezzo surface S' of degree at least five. The image of a k -point of S is a k -point of S' , so S' , and therefore S , is k -rational by [5, Theorem 2.1].

Finally, if S is stably k -rational, a nonempty open of a projective space is birational to a nonempty open of $S \times \mathbf{P}^m$. Since k is infinite, this supplies a k -point on S . Manin's theorem makes $\text{Pic}(S_{\bar{k}})$ stably permutation [2]. This proves the last assertion. $\square$

Proof of Corollary 1.3. The argument takes place entirely over the function field of the projection base. This avoids any specialization or fibrewise rationality claim.

For $X = X_{j,r}$, Tschinkel and Zhang project from a plane containing a distinguished line. The base is $\mathbf{P}_{\mathbf{Q}}^{r+1}$, with function field $K = \mathbf{Q}(t_1, \dots, t_{r+1})$, and the generic cubic-surface fibre is K -birational, after contracting the distinguished line, to a smooth quartic del Pezzo surface S/K with a K -point. Their calculation identifies the Galois image as type I_1 for $j = 1$ and type I_3 for $j = 3$ [4, Propositions 5.1–5.2]. Theorem 1.2 gives

$$K(S)(u, v) \simeq K(z_1, z_2, z_3, z_4).$$

Since $\mathbf{Q}(X) = K(S)$ as extensions of K ,

$$\mathbf{Q}(X)(u, v) \simeq \mathbf{Q}(t_1, \dots, t_{r+1}, z_1, z_2, z_3, z_4)$$

is purely transcendental. This is rationality of $X \times \mathbf{A}^2$, equivalently of $X \times \mathbf{P}^2$. The argument is an identity of function fields; no specialization assertion is used. $\square$

Proof of Theorem 1.1. The one-stabilization theorem states that, for every smooth complex cubic threefold X , the variety $X \times \mathbf{P}^1$ is irrational [3, Theorem 1.1]. It applies to $(X_j)_{\mathbf{C}}$. Therefore neither X_j nor $X_j \times \mathbf{P}^1$ can be rational over $\mathbf{Q}$, because rationality survives extension of the ground field. Corollary 1.3 gives rationality at $m = 2$ over $\mathbf{Q}$, and hence also over $\mathbf{C}$. This proves both equalities of stabilization levels. $\square$

Proof of Corollary 1.4. Theorem 1.1 makes $Y_j = X_j \times \mathbf{P}^1$ nonrational, even after base change to $\mathbf{C}$. On the other hand,

$$\mathbf{Q}(Y_j \times \mathbf{A}^1) = \mathbf{Q}(X_j)(u, t)$$

is purely transcendental by Corollary 1.3. Equivalently, $\mathbf{P}^1 \times \mathbf{A}^1$ is birational to $\mathbf{A}^2$. Hence $Y_j \times \mathbf{A}^1$ is rational. $\square$

Proof of Corollary 1.5. Theorem 1.1 gives the value two at the two displayed points. Engel, de Gaay Fortman, and Schreieder prove that a very general smooth complex cubic threefold does not admit a decomposition of the diagonal and hence is not stably rational [1, Corollary 1.4]. For such a threefold no finite m makes $X \times \mathbf{P}^m$ rational, so $\ell_{\mathbf{C}}(X) = \infty$. $\square$

A Exact tangent-section certificate

This appendix records the exact cover used in Proposition 3.2. It is included so that the geometric nonemptiness claim can be checked without treating the replay program as an oracle.

After discarding nonzero rational factors, the four evaluation determinants are

$$\begin{aligned}
D_1 &= (8a - 27)^2(3968a - 1349b - 2619), \\
D_2 &= 568453977a^3 - 373279764a^2b - 1658788404a^2 + 59068867ab^2 \\
&\quad + 976763788ab + 925668016a - 128537024b^2 - 359163728b - 10185728, \\
D_3 &= 1643918400a^3 - 1105797420a^2b - 7261914240a^2 + 154718524ab^2 \\
&\quad + 3804690316ab + 9944961600a - 282623593b^2 - 3443769343b - 3454184244, \\
D_4 &= 35068545a^3 + 2864538a^2b - 236376549a^2 - 5350827ab^2 \\
&\quad + 24417420ab + 469658574a + 14536855b^2 - 90750394b - 214068162,
\end{aligned}$$

and the corresponding smoothness minors are

$$\begin{aligned}
M_1 &= -4(a-1)(b-7)(3a-b-2)(2ab+7a-9b), \\
M_2 &= -9(a-1)(2b-11)(3a-b-2)(4ab+11a-15b), \\
M_3 &= -25(a-1)(3b-13)(2a-b-1)(3ab+13a-16b), \\
M_4 &= -(a-1)(4b-17)(13a-5b-8)(32ab+85a-117b).
\end{aligned}$$

To prove (14), suppose first that $D_i M_i = 0$ for $i = 1, 2, 3$. Choosing one factor from each product gives eight cases. A lexicographic Gröbner calculation in $\mathbf{Q}[a, b, h]/(h\Delta - 1)$ gives

chosen equations	localized zero locus
$D_1 D_2 D_3, D_1 D_2 M_3, D_1 M_2 D_3, D_1 M_2 M_3$	empty
$M_1 D_2 D_3, M_1 D_2 M_3$	empty
$M_1 M_2 D_3$	$3a - b - 2 = Q_0(b) = 0$
$M_1 M_2 M_3$	$3a - b - 2 = (3b - 26)(3b - 13) = 0$

where

$$Q_0(b) = 31223016b^2 - 435944529b + 1306078948.$$

On the line $3a - b - 2 = 0$, the primitive numerator of $D_4 M_4$ is

$$-(b-1)^4(4b-17)(16b-85)Q_4(b), \quad Q_4(b) = 83246b^2 - 872181b + 2185995.$$

The evaluations

$$\begin{aligned}
Q_0(17/4) &= \frac{69121705}{4}, & Q_0(85/16) &= -\frac{4117757269}{32}, \\
Q_4(26/3) &= \frac{7918133}{9}, & Q_4(13/3) &= -\frac{272530}{9}
\end{aligned}$$

exclude the four linear roots. Finally, the two quadratics are coprime by the identity

$$\begin{aligned}
&(2265746679974131615 - 377042650728395274b)Q_0(b) \\
&\quad + (141417109727495582904b - 1342668830289072756147)Q_4(b) \\
&= 24176690547344887359179755.
\end{aligned}$$

Thus $D_4 M_4 \neq 0$ in both remaining cases. This proves that the four opens $D_i M_i \neq 0$ cover $\Delta \neq 0$.

B Verification and scope

The cocharacter saturation, type- I_3 actions, Cox weights, determinant-one weight differences, and tangent-incidence cover use exact integer and rational arithmetic. The verification bundle checks all four pairs (D_i, M_i) , the eight localized branches, and the Bézout identity above with SymPy 1.14.0. A separate implementation checks the saturated lattice, the full type- I_3 action, the rank-two quotient lattice, and the displayed numerical tangent matrices. The quotient criterion, equivariant generic trivialization, incidence descent, and function-field argument are proved in the text and are not delegated to those programs.

No Lean development formalizes Theorem 1.1, Theorem 1.2, Theorem 2.1, Proposition 3.2, or the stated corollaries. The formal-coverage annotations therefore say **absent**.

The proof imports Tschinkel–Zhang’s universal-torsor geometry and their classification of the four hereditary- H^1 Galois images. The new input is the rational rank-three quotient and its descended tangent section. This is the step that lowers the number of stabilizing variables to two; no classification beyond those four Galois types is needed here.

AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author’s direction, the systems assisted with proof exploration, literature research, exact computation, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

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